

TM Forum Component

Resource Inventory

TMFC012

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Notice

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Direct inquiries to the TM Forum office:

100 Enterprise Drive
Suite 301 #1649
Rockaway, NJ 070866, USA
Tel No. +1 862 227 1648
TM Forum Web Page: www.tmforum.org

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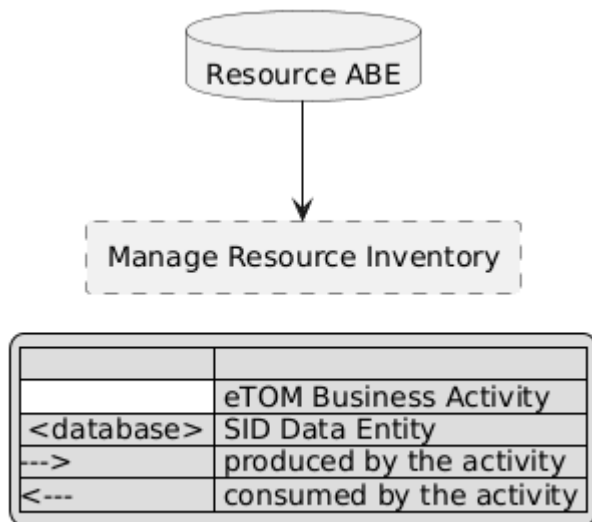
TMFC012 – Resource Inventory

1. Overview

Component Name	ID	Description	ODA Function Block
Resource Inventory	TMFC012	The Resource Inventory component owns accounting for resources (all inputs - including stock, parts, assets, production components etc.) that are owned and/or held for allocation and/or use by the organization. The Resource Inventory component has functionality that includes inventory item creation, inventory organization, inventory search or filter, inventory monitoring and tracking, inventory control (organization, re-stock management etc.) and inventory auditing. The minimum check to be performed at inventory item creation or update is for global consistency with related Resource Catalog information.	Production

2. eTOM Processes, SID Data Entities and Functional Framework Functions

2.1. eTOM L2 - SID ABEs links



2.2. eTOM business activities

eTOM business activities this ODA Component is responsible for:

Identifier	Level	Business Activity Name	Description
1.5.4	L2	Resource Readiness & Support	Manage resource infrastructure to ensure that appropriate application, computing and network resources are available and ready to support the Fulfillment, Assurance and Billing processes in instantiating and managing resource instances, and for monitoring and reporting on the capabilities and costs of the individual FAB processes.
1.5.4.5	L3	Manage Resource Inventory	Establish, manage and administer the enterprise's resource inventory, as embodied in the Resource Inventory Database, and monitor and report on the usage and access to the resource inventory, and the quality of the data maintained in it.

2.3. SID ABEs

SID ABEs this ODA Component is responsible for:

SID ABE L1	SID ABE L1 Definition	SID ABE L2 (or set of BEs)	SID ABE L2 Definition

Resource ABE	A Resource represents an instance of ResourceSpecification. The Resource ABE contains entities that are used to represent the three types of Resources: Logical, Physical and Compound Resources. The Resource carries the place where the Resource is installed, as well as configuration characteristics, such as assigned telephone numbers and internet addresses. The Resource also tracks the links to Product and/or Service realized.		(no description available)
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2.4. Functional Framework Functions

Function ID	Function Name	Aggregate Function L1	Aggregate Function L2	Function Description
426	Physical Implementation Information Capturing	Resource Management	Resource Repository Management	Physical Implementation Information Capturing provide levels of implementation details that tactical planning does not need to specify, such as duct routes and the frame appearances of device ports.
442	Network Overviews Presentation	Resource Management	Resource Repository Management	Network Overviews Presentation provides a more generalized view of the network than found in resource management.
453	Resource Configuration Change Logging	Resource Management	Resource Repository Management	Resource Configuration Change Logging - Collects and Records the history of configuration changes

454	Resource Configuration Management	Resource Management	Resource Repository Management	Resource Configuration Management provides configuration database and management of the configurations of the individual resources.
456	Resource Topology Verification	Resource Management	Resource Repository Management	Resource Topology Verification works with the Inventory Management functions to ensure that the topology reflected in its database is in sync with that in the Inventory Management Systems.
468	Resource Information Model Creation	Resource Management	Resource Repository Management	Resource Information Model creation function allows Service Providers to create information base in the resource inventory of the managed resources along its attributes. This function uses the standardized information model (e.g. TM Forum Information Framework) for the resources to be managed. The specific details will depend on the particular resources (e.g., particular types of managed elements and equipments) and associated technologies.

470	Resource Repository Retrieval	Resource Management	Resource Repository Management	This function allows for client operations support (service assurance and billing systems) to retrieve part or all of the resource inventory known to the target OSS. This feature may allow the following selection criteria: / • retrieval of a specified set of one or more sub-trees / • exclusion or inclusion of specified object types from the selected sub-tree / • further filtering based on attribute matching / • retrieval of only the object instances that have been modified after a provided date and time / • For the selected objects, this feature may allow the client operations support (service assurance and billing systems) to specify what specific attributes and relationships shall be returned. This (the attributes and relationships to be returned) would be the same for all objects of the same type.
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471	Resource Repository Updating	Resource Management	Resource Repository Management	Resource Repository Updating function entails update of the resource Repository based on a provided collection of updates. The expectation is that the Repository is updated as requested, but no other side effects are expected (e.g., creating a Sub Network Connection (SNC) in the network). This is a key point concerning this capability. The Repository update request can involve addition (new object), modification (change to an existing object) or deletion (removal of an object).
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472	Resource Repository Update Notification	Resource Management	Resource Repository Management	Resource Inventory Update Notification function entails the generation of inventory update notifications based on changes to the inventory: Notifications concerning object creation, object deletion and attribute value changes to other systems. / • Single Entity Notifications – in this variation of the feature, each notification pertains to only one entity, e.g., an equipment instance / • Multi-entity Notifications – in this variation of the feature, a single notification may report on inventory changes for multiple entities. / • Notification Suppression – in this variation of the feature, each notification pertains to only one entity.
562	Voucher Reporting	Resource Management	Resource Repository Management	Voucher Reporting function for querying and reporting of voucher related data
564	Voucher Life Cycle Management	Resource Management	Resource Repository Management	Voucher Life Cycle Management including activation, locking, expiration and maintenance of purchased vouchers.
738	Resource Data Inventory Synchronization	Resource Management	Resource Repository Management	Resource Data Inventory Synchronization is the function that ensure OSS Inventory data generated in each function is available to other functions as required.

436	Number Aging	Resource Management	Regulated Logical Resources Management	Manages the aging of numbers before they can be re-assigned.
437	Number Assigning	Resource Management	Regulated Logical Resources Management	Number Assigning manages the assignment of numbers for usage.
439	Number Searching	Resource Management	Regulated Logical Resources Management	Number Searching provides the means to search the number inventory.
440	Number Tracking and Reporting	Resource Management	Regulated Logical Resources Management	Provides functionality to track numbers and perform number reporting.
743	Number Portability Orchestration	Resource Management	Regulated Logical Resources Management	Number Portability Orchestration is a communication mechanism that ensures the Resource Number orders activation according to a criteria set, allowing in this way the correct execution of orders.
744	Number Portability Risk & Effectiveness Management	Resource Management	Regulated Logical Resources Management	Number Portability Risk & Effectiveness Management for determine of threats, risks and control of fulfillment, in order to comply with the execution of all Resource Number Portability orders.

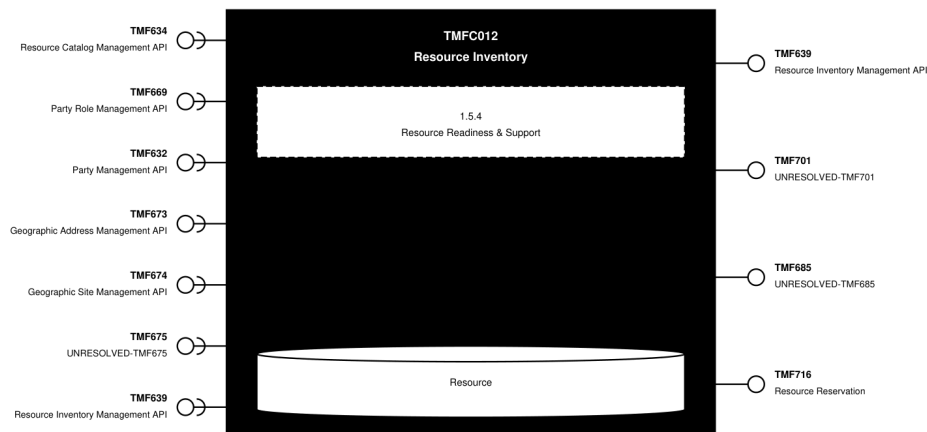
745	Number Portability Validation	Resource Management	Regulated Logical Resources Management	Number Portability Validation can perform calculations that determine whether the information received is reliable, safe and contains the minimum information required during implementation, enabling with this the Resource Number Portability orders rejection for those who do not comply with the defined criteria.
1062	Number Acquisition	Resource Management	Regulated Logical Resources Management	Number Acquisition manages the capturing of numbers for the number inventory.
1249	Number Porting	Resource Management	Regulated Logical Resources Management	Number Porting implements the changes to transfer the management of a number from one Service Provider to another, and on request provides status on the implementation of the changes.
438	Number Reservation	Resource Management	Regulated Logical Resources Management	Provides the functionality to manage the reservation of numbers.
435	Number Inventory Establishing	Resource Management	Regulated Logical Resources Management	Number Inventory Establishing manages the establishment of a number inventory.

3. TM Forum Open APIs & Events

The following part covers the APIs and Events; this part is split in 3: - List of Exposed APIs - This is the list of APIs available from this component. - List of Dependent APIs - In order to satisfy the provided API, the component could require the usage of this set of required APIs. - List of Events

(generated & consumed) - The events which the component may generate are listed in this section along with a list of the events which it may consume. Since there is a possibility of multiple sources and receivers for each defined event.

3.1. API Context Diagram

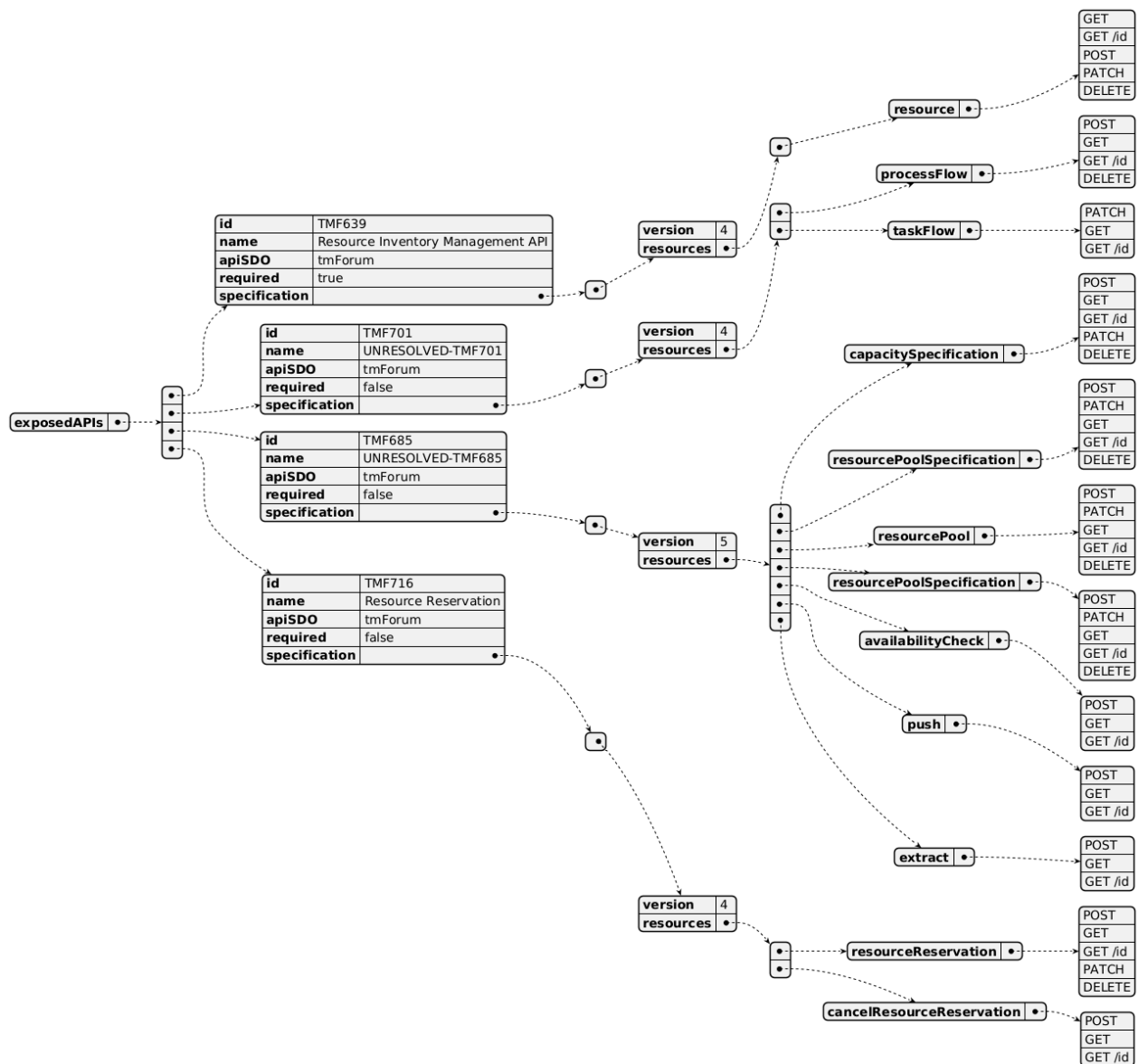


(SVG source: [TMFC012 API Context.svg](#) — hand-drawn rather than PlantUML for this one diagram, since PlantUML's automatic layout couldn't give each API its own straight, individually-anchored connector at a chosen height. Generated by the *component-specification-documentation skill's scripts/render_api_context_svg.py*.)

3.2. Exposed APIs

API ID	API Name	Mandatory / Optional	API Version	Resource	Operations
TMF639	Resource Inventory Management API	Mandatory	4	resource	GET, GET /id, POST, PATCH, DELETE
TMF701	UNRESOLVED-TMF701	Optional	4	processFlow	POST, GET, GET /id, DELETE
TMF701	UNRESOLVED-TMF701	Optional	4	taskFlow	PATCH, GET, GET /id
TMF685	UNRESOLVED-TMF685	Optional	5	capacitySpecification	POST, GET, GET /id, PATCH, DELETE
TMF685	UNRESOLVED-TMF685	Optional	5	resourcePoolSpecification	POST, PATCH, GET, GET /id, DELETE

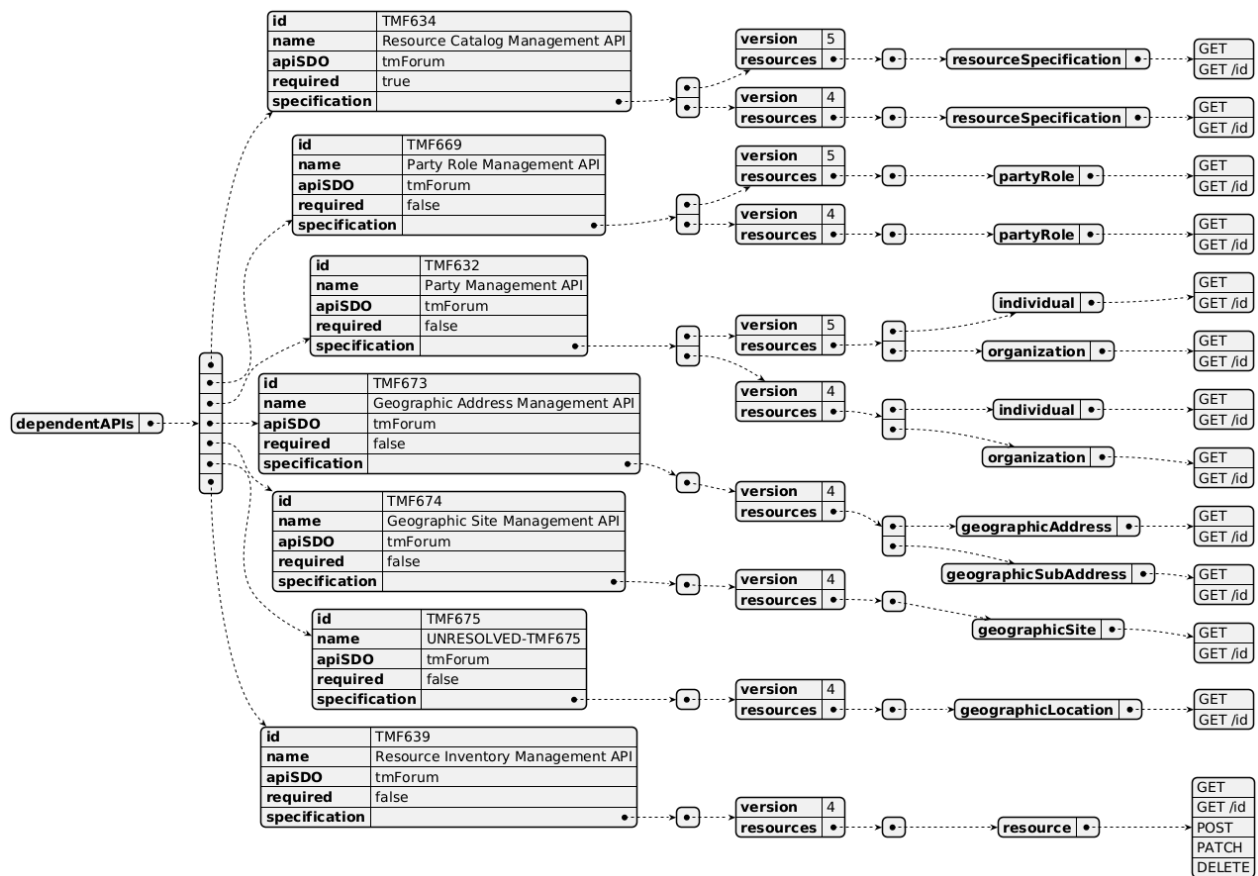
TMF685	UNRESOLVED-TMF685	Optional	5	resourcePool	POST, PATCH, GET, GET /id, DELETE
TMF685	UNRESOLVED-TMF685	Optional	5	resourcePoolSpecification	POST, PATCH, GET, GET /id, DELETE
TMF685	UNRESOLVED-TMF685	Optional	5	availabilityCheck	POST, GET, GET /id
TMF685	UNRESOLVED-TMF685	Optional	5	push	POST, GET, GET /id
TMF685	UNRESOLVED-TMF685	Optional	5	extract	POST, GET, GET /id
TMF716	Resource Reservation	Optional	4	resourceReservation	POST, GET, GET /id, PATCH, DELETE
TMF716	Resource Reservation	Optional	4	cancelResourceReservation	POST, GET, GET /id



3.3. Dependent APIs

API ID	API Name	API Version	Mandatory / Optional	Resource	Operations
TMF634	Resource Catalog Management API	5	Mandatory	resourceSpecification	GET, GET /id
TMF634	Resource Catalog Management API	4	Mandatory	resourceSpecification	GET, GET /id
TMF669	Party Role Management API	5	Optional	partyRole	GET, GET /id

TMF669	Party Role Management API	4	Optional	partyRole	GET, GET /id
TMF632	Party Management API	5	Optional	individual	GET, GET /id
TMF632	Party Management API	5	Optional	organization	GET, GET /id
TMF632	Party Management API	4	Optional	individual	GET, GET /id
TMF632	Party Management API	4	Optional	organization	GET, GET /id
TMF673	Geographic Address Management API	4	Optional	geographicAddress	GET, GET /id
TMF673	Geographic Address Management API	4	Optional	geographicSubAddress	GET, GET /id
TMF674	Geographic Site Management API	4	Optional	geographicSite	GET, GET /id
TMF675	UNRESOLVED-TMF675	4	Optional	geographicLocation	GET, GET /id
TMF639	Resource Inventory Management API	4	Optional	resource	GET, GET /id, POST, PATCH, DELETE



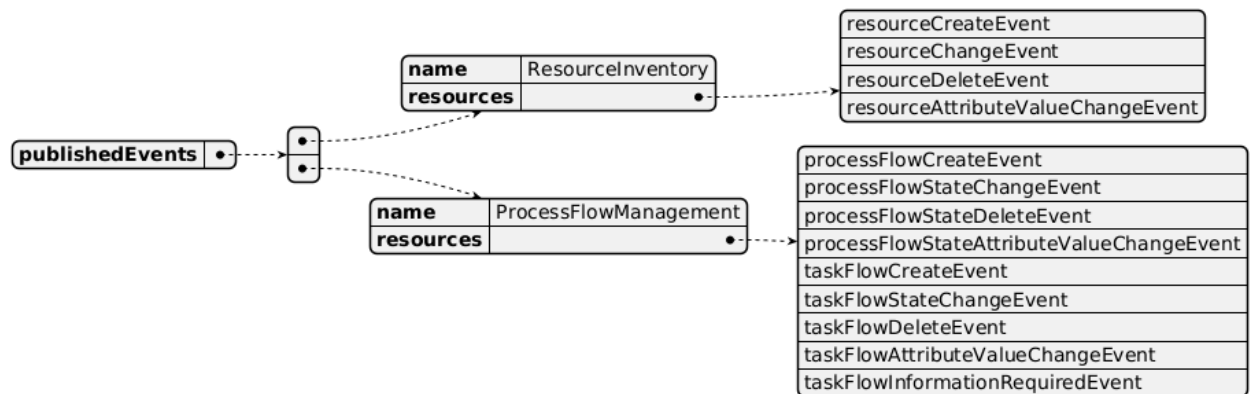
3.4. Events

The diagram illustrates the Events which the component may publish and the Events that the component may subscribe to and then may receive. Both lists are derived from the APIs listed in the preceding sections.

Published Events

API ID	API Name	Event Resources
TMF639	Resource Inventory Management API	resourceCreateEvent resourceChangeEvent resourceDeleteEvent resourceAttributeValueChangeEvent

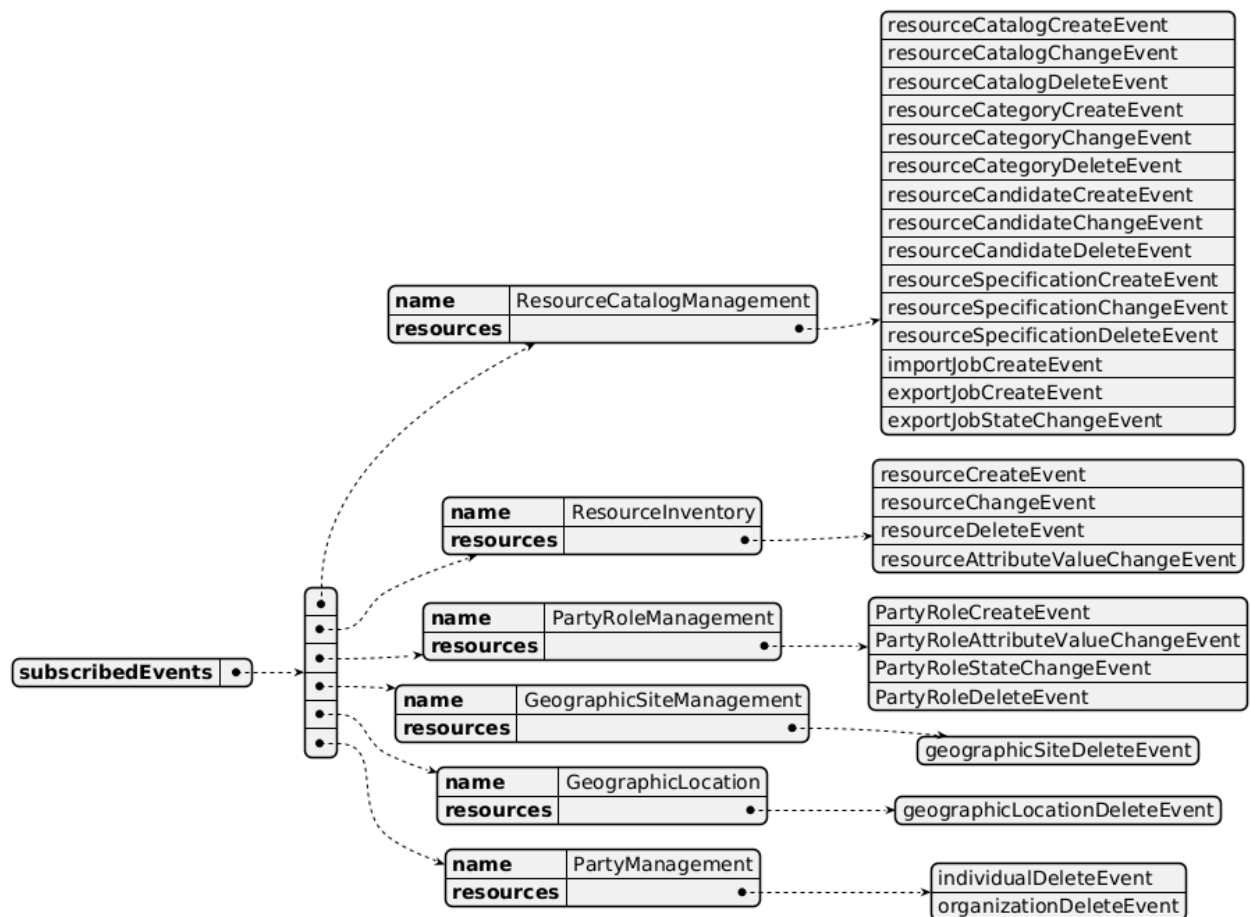
TMF701	UNRESOLVED-TMF701	processFlowCreateEvent processFlowStateChangeEvent processFlowStateDeleteEvent processFlowStateAttributeValueChangeEvent taskFlowCreateEvent taskFlowStateChangeEvent taskFlowDeleteEvent taskFlowAttributeValueChangeEvent taskFlowInformationRequiredEvent
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Subscribed Events

API ID	API Name	Event Resources
TMF634	Resource Catalog Management API	resourceCatalogCreateEvent resourceCatalogChangeEvent resourceCatalogDeleteEvent resourceCategoryCreateEvent resourceCategoryChangeEvent resourceCategoryDeleteEvent resourceCandidateCreateEvent resourceCandidateChangeEvent resourceCandidateDeleteEvent resourceSpecificationCreateEvent resourceSpecificationChangeEvent resourceSpecificationDeleteEvent importJobCreateEvent exportJobCreateEvent exportJobStateChangeEvent
TMF639	Resource Inventory Management API	resourceCreateEvent resourceChangeEvent resourceDeleteEvent resourceAttributeValueChangeEvent

TMF669	Party Role Management API	PartyRoleCreateEvent PartyRoleAttributeValueChangeEvent PartyRoleStateChangeEvent PartyRoleDeleteEvent
TMF674	Geographic Site Management API	geographicSiteDeleteEvent
TMF675	UNRESOLVED-TMF675	geographicLocationDeleteEvent
TMF632	Party Management API	individualDeleteEvent organizationDeleteEvent



4. Machine Readable Component Specification

Refer to the ODA Component Directory on the TM Forum website for the machine-readable component specification files for this component.

5. References

5.1. TMF Standards related versions

Standard	Version(s)
eTOM	v23.0.1
SID	v25.0
Functional Framework	v23.0

5.2. Jira References

Functional Framework - ISA-416 - Harmonize functions related to Product, Service and Resource Inventories (BACKLOG). Harmonize as much as possible functions related to Product, Service and Resource Inventory. - ISA-417 - Review Resource Inventory Mgt sub-domain functions (BACKLOG). Review functions 465, 466 and 564 in Resource Inventory Mgt sub-domains as they are not related to such an inventory.

API - <https://projects.tmforum.org/jira/browse/AP-1401>

ODA Components - TAC-342 - Create and Publish: TMFCxxx: Add a Resource Topology Mgt component to the component inventory (BACKLOG). Add a new Resource Topology Management component in the component inventory.

5.3. Further resources

This component is involved in the following use cases described in IG1228 How to use ODA - Using Open APIs to realize Use Cases.

6. Administrative Appendix

6.1. Document History

6.1.1. Version History

Version Number	Date	Modified by	Description of changes
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1.0.0	31-May-2022	Kamal Maghsoudlou, Sylvie Demarest	Initial release
1.0.1	25-Jul-2023	Ian Turkington	No content changed, simply a layout change to match template 3. Separated the YAML files to a managed repository.
1.0.1	14-Aug-2023	Amaia White	Final edits prior to publication
2.0.0	30-Apr-2024	Gaetano Biancardi	Updated Component Template; removed Resource Pool Management API from exposed list; aligned to Frameworks 23.5; removed TMF688 Event Management (from exposed and dependent API); removed TMF672 User Roles and Permissions from dependent API
2.0.0	12-Jul-2024	Amaia White	Final edits prior to publication
2.1.0	18-Nov-2024	Gaetano Biancardi	API version, only major version to be specified; Dependant API, following APIs moved from mandatory to optional (correction of diagram typo): TMF673, TMF674
2.1.0	24-Dec-2024	Rosie Wilson	Final edits prior to publication
2.1.1	28-Feb-2025	Rosie Wilson	Versioning updated ahead of update
2.1.1	09-Jul-2025	Rosie Wilson	Updates to description per agreed YAML updated on 03 Jun 2025
2.1.1	15-Jul-2025	Rosie Wilson	Final edits prior to publication
2.2.0	29-Jan-2026	julien rouland	Component Update: add exposed APIs TMF685 resourcePool; add exposed APIs TMF716 resource reservation; update functional framework (remove duplicated 471, update title and description of 472)
2.2.0	05-Jun-2026	Hugo Vaughan	Gen 5 API updates
2.2.0	25-Jun-2026	Sannah Sibaya	Final edits prior to publication

6.1.2. Release History

Release Status	Date Modified	Modified by	Description of changes
Pre-production	31-May-2022	Goutham Babu	Initial release
Production	22-Jul-2022	Adrienne Walcott	Updated to reflect TM Forum Approved status
Pre-production	14-Aug-2023	Amaia White	New release v1.0.1
Production	06-Oct-2023	Adrienne Walcott	Updated to reflect TM Forum Approved status
Pre-production	12-Jul-2024	Amaia White	New release v2.0.0
Production	30-Aug-2024	Adrienne Walcott	Updated to reflect TM Forum Approved status
Pre-production	24-Dec-2024	Rosie Wilson	New release v2.1.0
Production	07-Mar-2025	Adrienne Walcott	Updated to reflect TM Forum Approved status
Pre-production	15-Jul-2025	Rosie Wilson	New release v2.1.1
Production	05-Sep-2025	Adrienne Walcott	Updated to reflect TM Forum Approved status
Pre-production	25-Jun-2026	Sannah Sibaya	New release v2.2.0

6.2. Acknowledgements

This document was prepared by the members of the TM Forum ODA Components & Canvas team.

Team Member	Company	Role
Kamal Maghsoudlou	Ericsson	Editor
Sylvie Demarest	Orange	Reviewer
Hugo Vaughan (TM Forum)	TM Forum	Additional Input

Ian Turkington	TM Forum	Additional Input
Gaetano Biancardi	Accenture	Additional Input
Abinash Vishwakarma	Netcracker	Reviewer
Jose Macaluso	Xacria	Reviewer
Milind Bhagwat	BT	Reviewer
Ritu Arora	BT	Reviewer
Matteo Destino	Accenture	Reviewer
Elisabeth Andersson	MATRIX	Reviewer